Question ID: 86684ce9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The result of increasing the quantity x by $1,800\%$ is 684. What is the value of x?

Answer

A. $12,996$

B. $12,312$

C. 38

D. 36

Correct Answer: D

Rationale

Choice D is correct. It's given that the result of increasing the quantity x by $1,800\%$ is 684. It follows that $x + \left(\frac{1,800}{100}\right)x = 684$, which is equivalent to $x + 18x = 684$, or $19x = 684$. Dividing each side of this equation by 19 yields $x = 36$. Therefore, the value of x is 36.

Choice A is incorrect. The result of increasing the quantity $12,996$ by $1,800\%$ is $246,924$, not 684.

Choice B is incorrect. The result of increasing the quantity $12,312$ by $1,800\%$ is $233,928$, not 684.

Choice C is incorrect. The result of increasing the quantity 38 by $1,800\%$ is 722, not 684.